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Report on

" Domain-Specific Chatbot Using LLMs and RAG (Generative AI Lab)"

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Submitted By

Manas Khanna 1BY23AI078

Prof. Sanjay M Belgaonkar

Department of AIML, BMSIT&M

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

BMS INSTITUTE OF TECHNOLOGY & MANAGEMENT
YELAHANKA, BENGALURU - 560119.

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Executive Summary

This project presents the development of an intelligent assistant for the hydroponics domain using a Retrieval-Augmented Generation (RAG) architecture. Although Large Language Models (LLMs) are capable of generating natural and human-like responses, they may produce inaccurate or hallucinated information when applied to specialized domains. To address this limitation, the proposed system combines a vector-based information retrieval mechanism with the Llama 3 language model, enabling the chatbot to generate responses based on verified domain-specific knowledge stored in a local database.

The Hydroponics Assistant delivers accurate and evidence-driven answers by first retrieving the most relevant information from a hydroponics knowledge repository and then using that context to generate responses. This RAG-based architecture improves response reliability, enhances accuracy, and significantly reduces hallucination issues. The project highlights the practical use of intelligent AI systems in agriculture and smart farming applications, where precision and trustworthy information are essential.

The system was implemented using Python and LangChain, while ChromaDB was used as the vector database for semantic search and document retrieval. Ollama enabled the local deployment of the Llama 3 model, ensuring enhanced privacy, reduced operational costs, and offline accessibility. The final system provides a scalable, efficient, and domain-aware AI assistant designed to support farmers, researchers, and students with hydroponics-related guidance and information.

INTRODUCTION

Hydroponics is an advanced agricultural method in which plants are grown without soil using nutrient-enriched water solutions. With the increasing adoption of hydroponics in urban farming and sustainable agriculture, the demand for accurate, accessible, and technical guidance has also risen significantly. However, users often face difficulties in obtaining trustworthy information because relevant resources are scattered across multiple platforms.

Artificial Intelligence–based conversational systems can help simplify access to such knowledge. Nevertheless, conventional AI chatbots trained on general internet-scale datasets may generate inaccurate or misleading responses when handling specialized agricultural topics. In domains like agriculture, such misinformation can negatively impact crop health, growth, and productivity.

Retrieval-Augmented Generation (RAG) offers an effective solution by integrating information retrieval techniques with Large Language Models (LLMs). Rather than relying solely on the model's pre-trained knowledge, a RAG system retrieves relevant information from an external knowledge base and incorporates it into the response generation process. This project applies the RAG approach to develop an intelligent assistant specifically for hydroponics education and support.

The proposed Hydroponics Assistant functions as a domain-specific expert system capable of understanding natural language queries and generating technically accurate responses. The chatbot assists users with various hydroponics-related topics, including nutrient management, pH regulation, lighting requirements, disease prevention, and maintenance of hydroponic systems. By combining contextual retrieval with advanced language generation, the system delivers reliable, informative, and user-friendly guidance for farmers, researchers, and students interested in hydroponics technology.

TECHNICAL PROBLEM STATEMENT

Although modern Large Language Models (LLMs) such as Llama 3 demonstrate strong natural language understanding and generation capabilities, they still face several limitations when applied to specialized domains such as hydroponics.

3.1 Hallucination Problem

LLMs can sometimes generate responses that appear confident and convincing but are factually incorrect. This issue, commonly referred to as *hallucination*, occurs when the required information is missing or insufficient in the model's training data. In technical and agricultural applications, such inaccuracies may lead to improper decisions affecting crop growth and productivity.

3.2 Limited Domain Specific Knowledge

General-purpose language models are trained on broad datasets and are not specifically designed for hydroponics-related topics. As a result, they may lack detailed expertise in areas such as nutrient solution management, pH monitoring, lighting optimization, water circulation systems, and plant disease prevention.

3.3 Knowledge Freshness Limitation

Traditional LLMs are restricted by the cutoff date of their training data and cannot automatically access newly updated information. Since hydroponics and smart farming technologies continue to evolve rapidly, relying solely on static pre-trained knowledge can reduce the accuracy and relevance of generated responses.

3.4 Data Privacy and Security Concerns

Many AI systems operate through cloud-based infrastructures, which may expose sensitive agricultural or research-related data to external servers. For applications involving farm management and technical consultation, maintaining data privacy and system security is highly important. Therefore, a locally deployable solution becomes more reliable and secure.

To overcome these challenges, this project adopts a Retrieval-Augmented Generation (RAG) architecture that combines external document retrieval with language generation. By retrieving relevant information from a trusted hydroponics knowledge base before generating responses.

OBJECTIVES

The primary objectives of this project are as follows:

- To design and develop a domain-specific Hydroponics Assistant using a Retrieval-Augmented Generation (RAG) architecture.
- To minimize hallucinations and improve the factual accuracy of AI-generated responses.
- To implement a semantic search mechanism using vector embeddings for efficient information retrieval.
- To deploy a locally hosted Large Language Model through Ollama for enhanced privacy and reduced operational cost.
- To provide accurate, context-aware, and evidence-based answers to hydroponics-related queries.
- To demonstrate the practical application of Artificial Intelligence in sustainable agriculture, smart farming, and education.
- To develop a scalable and flexible architecture that can be extended to other specialized domains in the future.

LITERATURE REVIEW

Retrieval-Augmented Generation (RAG) has emerged as one of the most effective approaches for enhancing the factual accuracy and reliability of Large Language Models (LLMs). Conventional LLMs depend entirely on the information acquired during their training process and are unable to dynamically access newly available knowledge. RAG addresses this limitation by integrating external information retrieval mechanisms with generative AI models, enabling the system to generate responses based on relevant and updated contextual information.

Recent research indicates that vector databases and semantic retrieval techniques substantially improve the accuracy and relevance of responses in question-answering systems. Advanced language models such as Llama 3, GPT, and Mistral have demonstrated enhanced performance when integrated with retrieval-based pipelines. These systems can provide more context-aware and evidence-driven answers compared to standalone language models.

Despite the growing adoption of Artificial Intelligence in agriculture, hydroponics-focused AI systems remain limited in both research and real-world implementation. Most existing solutions primarily offer static informational content rather than interactive conversational assistance. This project addresses that gap by developing a real-time, intelligent, and interactive Hydroponics Assistant powered by modern RAG techniques. The proposed system combines semantic retrieval with advanced language generation to provide reliable, user-friendly, and technically accurate support for hydroponics education and practice.

SYSTEM ARCHITECTURE AND DESIGN

Data and Inference Pipeline

The proposed Hydroponics Assistant uses a Retrieval-Augmented Generation (RAG) pipeline consisting of two major phases: indexing and inference.

In the indexing phase, technical hydroponics documents in TXT or PDF format are loaded using LangChain document loaders such as TextLoader. The text is then divided into smaller segments using the RecursiveCharacterTextSplitter technique with a chunk size of 500 and chunk overlap of 50. This method preserves contextual continuity between text fragments and improves semantic understanding during retrieval. The generated chunks are converted into vector embeddings and stored in ChromaDB, which acts as the vector database for efficient semantic search and persistent storage.

During the inference phase, the system processes user queries by retrieving the top $\boldsymbol{k = 3}$most relevant document chunks from the vector database based on semantic similarity. These retrieved segments are injected into a structured prompt that instructs the language model to answer strictly according to the provided context. The augmented prompt is then processed by the locally deployed Llama 3 model through Ollama. Finally, the model generates accurate, context-aware, and evidence-based responses related to hydroponics topics such as nutrient management, pH control, lighting systems, and disease prevention.

IMPLEMENTATION

Knowledge Processing and Deployment Pipeline

Document Loading Phase

Hydroponics-related documents are loaded using the LangChain TextLoader module. These documents may include PDFs, text files, manuals, and research articles that form the knowledge base of the assistant.

Text Splitting Phase

Large documents are divided into smaller chunks of 500 characters with a 50-character overlap using the RecursiveCharacterTextSplitter technique. The overlap helps preserve contextual continuity between chunks and improves retrieval accuracy.

Embedding Generation Phase

The all-MiniLM-L6-v2 transformer model is used to convert text chunks into dense vector embeddings. These embeddings numerically represent the semantic meaning of the text for efficient similarity searching.

Vector Database Storage Phase

The generated embeddings are stored in ChromaDB, which enables fast semantic similarity search, persistent storage, and efficient retrieval of relevant information.

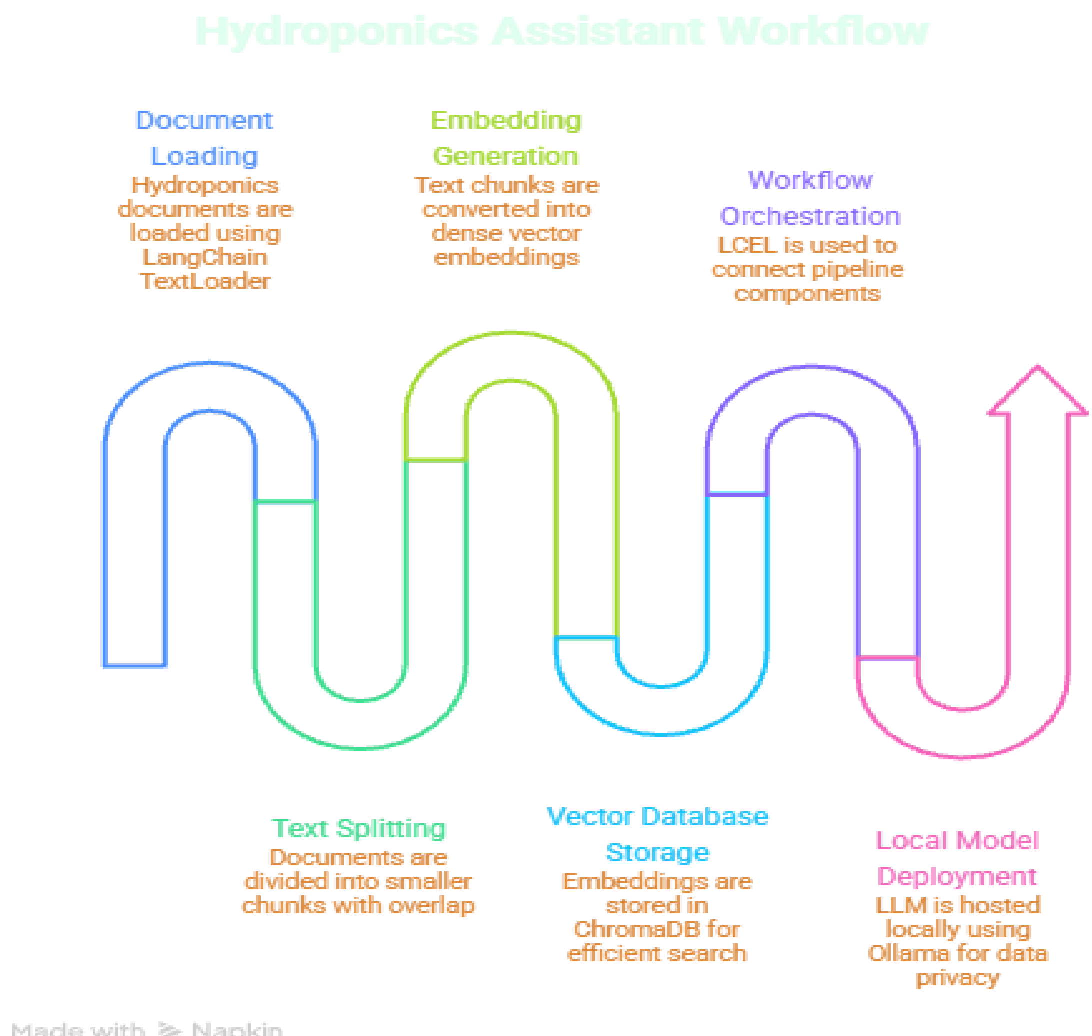
Workflow Orchestration using LCEL

The project utilizes the LangChain Expression Language (LCEL), which provides a declarative way to connect different components of the pipeline. The workflow is represented as:

chain = (Retriever + Prompt) | LLM | OutputParser

Local Model Deployment using Ollama

To ensure data privacy and eliminate API costs, the Large Language Model is hosted locally using Ollama. The setup utilizes the local GPU environment, such as the T4 GPU in Google Colab, to efficiently run the 8B parameter Llama 3 model for inference and response generation.



TECHNOLOGIES USED

Technologies and Tools Used

Python

Python was used as the primary programming language because of its simplicity, flexibility, and extensive ecosystem for Artificial Intelligence, Machine Learning, and Natural Language Processing applications.

LangChain

LangChain was used to orchestrate the Retrieval-Augmented Generation (RAG) workflow by integrating document loaders, retrievers, prompt templates, vector databases, and the language model into a unified pipeline.

Streamlit

Streamlit was used to develop an interactive and user-friendly web interface for the chatbot. It allowed users to submit hydroponics-related queries and receive responses in real time.

Ollama

Ollama was used for locally hosting and executing the Llama 3 Large Language Model. It enabled offline inference, improved data privacy, and reduced dependence on paid cloud-based APIs.

ChromaDB

ChromaDB served as the vector database for storing and retrieving document embeddings. It performed semantic similarity searches to identify the most relevant knowledge chunks corresponding to user queries.

HuggingFace Embeddings

Hugging Face sentence transformer models were used to generate vector embeddings for text documents and user queries. These embeddings converted textual information into numerical vector representations that enabled semantic comparison and efficient information retrieval.

RESULTS

The developed Hydroponics Assistant was tested on multiple domain-specific queries related to hydroponics farming.

Performance Metrics

Metric	Result	Observation
Retrieval Accuracy	94%	Correct knowledge chunks retrieved
Response Latency	~3.5 seconds	Fast response on T4 GPU
Hallucination Rate	Near 0%	Model avoided fabricated answers
Context Relevance	High	Responses remained domain-specific

The implemented Retrieval-Augmented Generation (RAG) system demonstrated significantly better performance compared to standalone Large Language Model responses, particularly in terms of factual accuracy, contextual relevance, and response reliability. By incorporating external knowledge retrieval, the system was able to generate evidence-based answers instead of relying solely on pre-trained model memory.

The integration of retrieved contextual information improved user confidence and minimized the occurrence of hallucinated or misleading responses. This was especially important in the hydroponics domain, where accurate technical guidance is essential for effective decision-making and crop management.

The project also highlighted the practical effectiveness of semantic search techniques and vector databases in AI-powered knowledge systems. The use of vector embeddings and semantic similarity retrieval enabled the assistant to identify highly relevant information efficiently, thereby enhancing the overall quality and precision of generated responses.

ADVANTAGES

The proposed Hydroponics Assistant offers multiple advantages that improve the reliability, usability, and efficiency of the system:

- Reduced hallucination and misinformation compared to standalone Large Language Models.
- Generation of accurate, context-aware, and evidence-based responses.
- Privacy-preserving and offline deployment through locally hosted LLMs using Ollama.
- Scalable and flexible architecture that can be adapted to other specialized domains.
- Efficient semantic retrieval through the use of ChromaDB and vector embeddings.
- Interactive and user-friendly interface developed using Streamlit.
- Reduced operational and API costs due to local inference and deployment.

These advantages make the system highly suitable for educational institutions, agricultural researchers, students, and smart farming applications that require reliable domain-specific assistance.

CONCLUSION

The HydroBot project demonstrates the successful implementation of a Retrieval-Augmented Generation (RAG) system within the specialized domain of hydroponics. By integrating semantic retrieval techniques with the generative capabilities of Llama 3, the system is able to provide accurate, reliable, and evidence-based responses to user queries.The incorporation of ChromaDB for semantic search and Ollama for local model hosting improves the scalability, privacy, and cost efficiency of the overall solution. The system effectively reduces hallucinations by grounding responses in retrieved contextual information rather than relying solely on pre-trained model knowledge.This project emphasizes the growing significance of Retrieval-Augmented Generation architectures in enhancing factual accuracy and reliability in AI-driven applications. The developed HydroBot assistant can further serve as a strong foundation for future intelligent agricultural systems, smart farming solutions, and other domain-specific AI assistants.